

CS1231S: Discrete Structures
Tutorial #4: Relations & Equivalence Relations
(Week 6: 19 – 23 Feb 2024)

1. Discussion Questions

These are meant for you to discuss on Canvas or QnA. No answers will be provided.

D1. Let $A = \{0,1\}$, $B = \{a,b,c\}$ and $C = \{01,10\}$. Determine the following:

- (a) $B \times C$ (b) $A \times B \times C$ (c) $\emptyset \times A$ (d) $\mathcal{P}(\{\emptyset\}) \times A$

D2. (AY2022/23 semester 1 midterm test)

Let R and S be binary relations on $A = \{a,b,c\}$ as follows:

$$R = \{(a,a), (a,b), (b,c), (c,b)\};$$

$$S = \{(a,a), (b,b), (c,a), (c,b)\}.$$

State whether the following statements are true or false.

- (a) $|S \circ R| = |R \circ S|$.
(b) $|R \circ R| = |S \circ S|$.
(c) $|(S \circ R) \circ S| = |S \circ (R \circ S)|$.

2. Tutorial Questions

1. Let $A = \{1,2, \dots, 10\}$ and $B = \{2,4,6,8,10,12,14\}$. Define a relation R from A to B by setting

$$x R y \Leftrightarrow x \text{ is prime and } x \mid y$$

for each $x \in A$ and each $y \in B$. Write down the sets R and R^{-1} in **roster notation**. Do not use ellipses (...) in your answers.

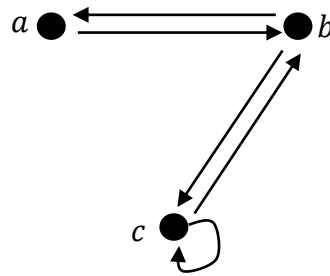
2. Let R be a relation on a set A . Show that the following are logically equivalent by using this strategy: (a) implies (b), (b) implies (c), and (c) implies (a).

- (a) R is symmetric, i.e. $\forall x, y \in A (x R y \Rightarrow y R x)$.
(b) $\forall x, y \in A (x R y \Leftrightarrow y R x)$.
(c) $R = R^{-1}$.

3. For each of the relations defined below, determine whether it is (i) reflexive, (ii) symmetric, (iii) transitive, and (iv) an equivalence relation. If a property is false for the relation, give a counter-example.

- (a) Let $A = \{1,2,3\}$, $Q = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$, where Q is a relation on A .
(b) Define the relation E on $\mathbb{Q}$ by setting, for all $x, y \in \mathbb{Q}$, $x E y \Leftrightarrow x = y$.
(c) Define the relation R on $\mathbb{Q}$ by setting, for all $x, y \in \mathbb{Q}$, $x R y \Leftrightarrow xy \geq 0$.
(d) Define the relation S on $\mathbb{Q}$ by setting, for all $x, y \in \mathbb{Q}$, $x S y \Leftrightarrow xy > 0$.
(e) Define the relation T on $\mathbb{Z}$ by setting, for all $x, y \in \mathbb{Z}$, $x T y \Leftrightarrow -2 \leq x - y \leq 2$.

4. The directed graph of a binary relation R on a set $A = \{a, b, c\}$ is shown below.



Draw the directed graph for each of the following and determine if it is transitive or not. If it is not transitive, explain.

- (a) $R \circ R$ (b) $R \circ R \circ R$ (c) $(R \circ R) \cup R$

5. Consider the relation $S = \{(m, n) \in \mathbb{Z}^2 : m^3 + n^3 \text{ is even}\}$. (Recall that $\mathbb{Z}^2$ means $\mathbb{Z} \times \mathbb{Z}$.) Determine (a) S^{-1} , (b) $S \circ S$, and (c) $S \circ S^{-1}$.

You may use theorems involving the sum of even and odd integers without quoting them (for example: the sum of two even integers is even; the sum of an even integer and odd integer is odd; etc.).

6. Let A, B, C, D be sets and $R \subseteq A \times B, S \subseteq B \times C$, and $T \subseteq C \times D$. Prove that

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

That is, composition of relations is associative.

7. (AY2020/21 semester 1 exam question)

Define an equivalence relation $\sim$ on $\mathbb{Z}^+ \times \mathbb{Z}^+$ by setting, for all $a, b, c, d \in \mathbb{Z}^+$,

$$(a, b) \sim (c, d) \Leftrightarrow ab = cd.$$

Write down the equivalence classes $[(1, 1)]$ and $[(4, 3)]$ in **roster notation**.

8. Define a relation $\sim$ on $\mathbb{Z} \setminus \{0\}$ as follows: $\forall a, b \in \mathbb{Z} \setminus \{0\} (a \sim b \Leftrightarrow ab > 0)$.

- (a) Prove that $\sim$ is an equivalence relation. You may adopt the appropriate **order axioms** and **theorems** in *Appendix A: Properties of the Real Numbers* for the integers. (Appendix A is available on Canvas > Files as well as the CS1231S webpage at https://www.comp.nus.edu.sg/~cs1231s/2_resources/lectures.html.)

- (b) Determine all the distinct equivalence classes formed by this relation $\sim$.

9. Let $\mathcal{C}$ be a partition of a set A . Denote by $\sim$ the same-component relation with respect to $\mathcal{C}$, i.e. for all $x, y \in A$,

$$x \sim y \Leftrightarrow x \text{ is in the same component of } \mathcal{C} \text{ as } y.$$

$$\Leftrightarrow x, y \in S \text{ for some } S \in \mathcal{C}.$$

- (a) Prove that if $x \in S \in \mathcal{C}$, then $[x] = S$.

- (b) Prove that $A/\sim = \mathcal{C}$. (Recall that $A/\sim$ denotes the set of all equivalence classes w.r.t. $\sim$)